

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	6 February 2026
Team ID	LTVIP2026TMIDS24218
Project Name	Civil Engineering Insight Studio
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable includes the architectural diagram as below and the information as per the table1 & table 2

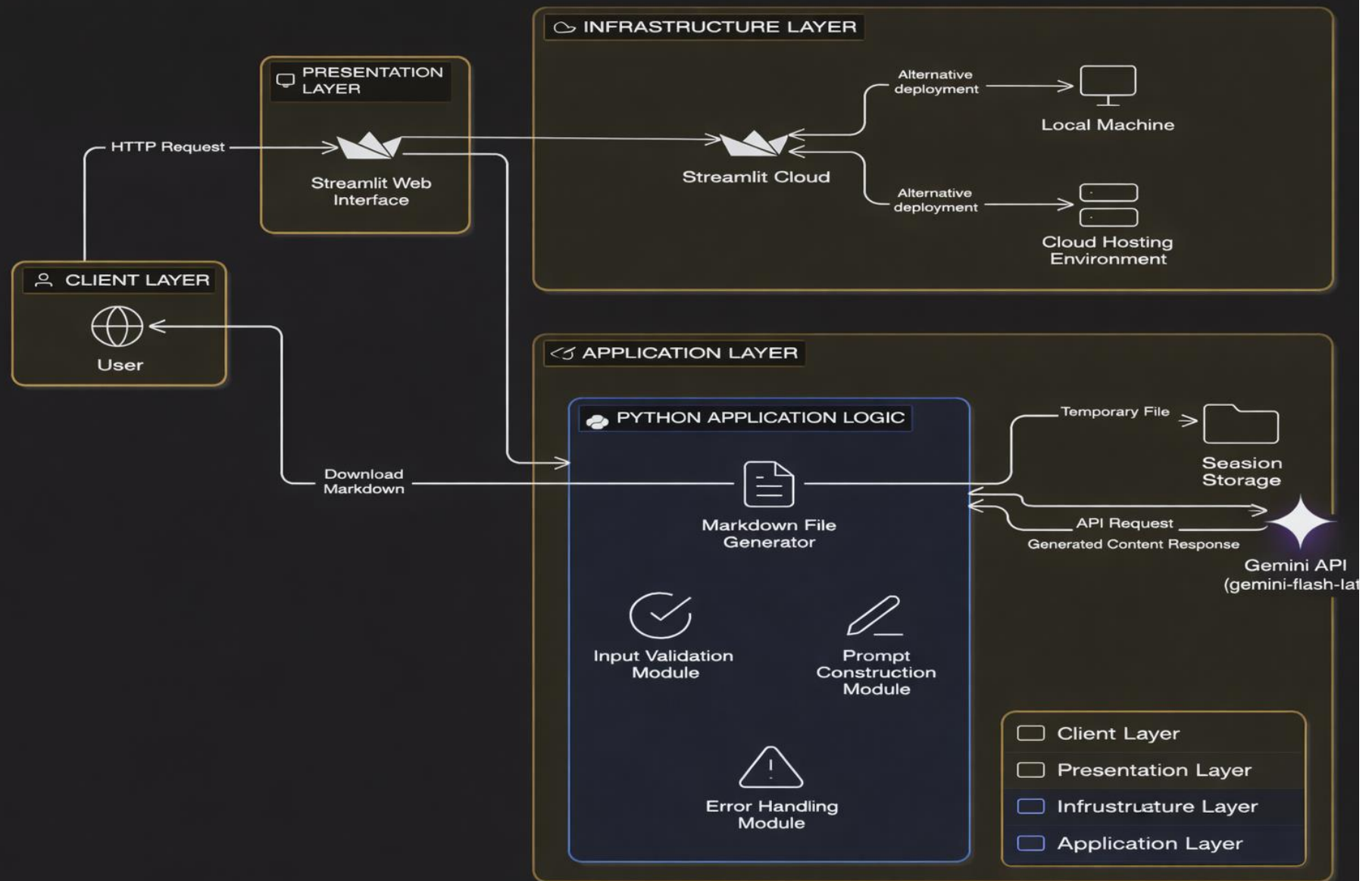


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based interface where users upload construction images, enter analysis request, and view generated structural report.	Streamlit (Python-based Web UI)
2.	Application Logic-1	Handles image upload, user input validation, and workflow control.	Python
3.	Application Logic-2	Prompt construction logic for structured civil engineering analysis.	Python (Prompt Engineering Logic)
4.	Application Logic-3	AI-powered structural analysis via external model integration.	Google Gemini API (gemini-flash-latest)
5.	Database	No persistent database used (stateless application).	Not Applicable
6.	Cloud Database	Not used in current implementation	Not Applicable
7.	File Storage	Temporary storage of generated report during session runtime.	Local Runtime (Streamlit Session State)
8.	External API-1	AI-based multimodal analysis (Image + Text).	Google Gemini API
9.	External API-2	Not used	Not Applicable
10.	Machine Learning Model	Large Multimodal Language Model for structural description and analysis.	Gemini Vision / Gemini Flash Model
11.	Infrastructure (Server / Cloud)	Application can be deployed locally or on cloud hosting platforms.	Local System / Streamlit Cloud / Python-supported Cloud Environment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Framework used to build and deploy the web interface	Streamlit (Open-source Python framework)
2.	Security Implementations	API key secured using environment variables (.env) or Streamlit secrets; no sensitive user data stored.	python-dotenv / Streamlit Secrets
3.	Scalable Architecture	Stateless architecture; scalability depends on hosting environment and Gemini API limits.	Streamlit + Cloud Hosting
4.	Availability	Application availability depends on hosting platform uptime.	Streamlit Cloud / Cloud Hosting Provider
5.	Performance	AI analysis response time depends on Gemini API latency; lightweight UI ensures minimal frontend delay.	Python + Gemini API